535514 Reinforcement Learning

Pre-Lecture Assignment: Lecture 10

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Problem: Distributional RL

In Lecture 9, we discussed multiple value-based RL algorithms, including the classic Q-learning. Notably, these methods all focus on learning the "expected" Q values. In our next lecture, we will go ahead to learn how to learn the distributions behind the Q functions. Let's first have warm-up practice before the lecture. You can refer to the short lecture video by Pascal Poupart: **https://www.youtube.com/watch?v=r-Yk6-jagDU** as a starting point when answering the questions below.

1) **Return as a random variable**: Let $Z^\pi(s,a) := \sum_{t=0}^{\infty} \gamma^t r_{t+1}$ under $s_0 = s, a_0 = a$ and a policy π.

Clearly, Z is a random variable. Can you write down the connection between $Z^\pi(s,a)$ and $Q^\pi(s,a)$?

2) **Distributional Bellman equation**: We know the Bellman expectation equation suggests that $Q^\pi(s,a) = R_s^a + \gamma \mathbb{E}_{s' \sim P(\cdot|s,a), a' \sim \pi(\cdot|s')}[Q^\pi(s',a')]$. Can you write down the "distributional" version of this Bellman equation?

3) **Distributional Bellman operator**: Based on 2), can you define a Bellman operator? Moreover, what nice properties can you get from this distributional Bellman operator?

Submission Guidelines and Remarks

- Your deliverable shall be a PDF file. Please put all your write-ups into one PDF file (photos/scanned copies are acceptable) and submit your deliverable via E3.
- Please submit it by 9pm, 4/21 (Monday).
- If you have any questions, feel free to post them on Slack.